Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025

Course: B.Tech
Code: 100202

Subject: Engineering Mathematics - II

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) If $f(z) = u(x, y) + i\,v(x, y)$ is analytic function, then $f'(z) =$

(i) $\frac{u}{x} - i\frac{v}{x}$
(ii) $\frac{u}{x} - i\frac{v}{y}$
(iii) $\frac{v}{y} - i\frac{v}{x}$
(iv) $\frac{u}{x} - i\frac{v}{y}$

(b) For an analytic function $f(z) = u(x, y) + i\,v(x, y)$, if the real part $u(x, y) = x^2 - y^2$ is given, what is the imaginary part $v(x, y)$

(i) $2xy + c$
(ii) $xy + c$
(iii) $x^2 + y^2 = c$
(iv) $-2xy + c$

(c) The value of the integral $\oint c\,|z|dz$, where C is the left half of theunit circle |z| = 1 from z = -I toz = i

(i) $2pi$
(ii) pi
(iii) 0
(iv) $2i$

(d) The residue of $e^z / \cos \pi z$ at its pole $z = -1/2$ is

(i) $-e^{1/2} / \pi$
(ii) $e^{1/2} / \pi$
(iii) 0
(iv) $-e^{1/2}$

(e) The particular integral y_p of the differential equation $(D^2 + 4)y = \cos 2x$, $D \equiv \frac{d}{dy}$ is

(i) $\frac{1}{2}\sin 2x$
(ii) $\frac{1}{2}x\sin 2x$
(iii) $\frac{1}{4}x\sin 2x$
(iv) $\frac{1}{2}x\cos 2x$

(f) By the method of undetermined coefficients, the particular integral y_p of the differential equation $(D^4 + 2D^3 + D^2)y = 12x^2$, $D \equiv \frac{d}{dy}$ is

(i) $a + bx + cx^2$
(ii) $ax + bx^2 + cx^3$
(iii) $x(a + bx + cx^2)$
(iv) $x^2(a + bx + cx^2)$

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(g) Let $\lim_{n\to\infty}(u_n)^{1/n} = \ell$, then the series $\sum_{n=1}^{\infty} u_n$ converges if

(i) $\ell > 1$ (iii) $\ell = 1$

(ii) $\ell < 1$ (iv) $\ell \geq 1$

(h) $L(\sin at) = \dfrac{a}{s^2 + a^2}$ when

(i) $s > a$ (iii) $s < 0$

(ii) $s < a$ (iv) $s > 0$

(i) If $L^{-1}\left(\dfrac{3s+8}{s^2+4s+25}\right) = e^{kt}\left[3\cos\sqrt{21}\,t + \dfrac{2}{\sqrt{21}}\sin\sqrt{21}\,t\right]$ then the value of k is

(i) -1 (iii) -2

(ii) 1 (iv) 2

(j) If $f(x) = \sqrt{\dfrac{1-\cos x}{2}}, 0 < x < 2\pi$ then the value of a_0 is

(i) $\dfrac{4}{\pi}$ (iii) $\dfrac{\pi}{4}$

(ii) $\dfrac{2}{\pi}$ (iv) $\dfrac{\pi}{2}$

Q.2 (a) Find $\lim_{z\to 0}\left[\dfrac{1}{1-e^{1/x}} + i y^2\right]$, if it exists. **[3]**

(b) Check the continuity of the following function at $z = 0$: **[3]**

$$f(z) = \begin{cases} \dfrac{\mathrm{Re}(z^2)}{|z|^2}, & z \neq 0 \\ 0, & z = 0. \end{cases}$$

(c) Examine whether the Cauchy-Riemann equations are satisfied at the origin for the function given below or not. **[8]**

$$f(z) = \begin{cases} \dfrac{x^3(1+i) - y^3(1-i)}{x^2+y^2}, & z \neq 0 \\ 0, & z = 0. \end{cases}$$

Also check, whether $f(z)$ is analytic at $z = 0$ or not.

Q.3 (a) If $f(\xi) = \int_C \dfrac{3z^2+7z+1}{z-\xi}\,dz$, where C is an ellipse $9x^2 + 16y^2 = 144$. Find the values of $f(3)$, $f'(1-i)$, and $f''(1-i)$. **[6]**

(b) Evaluate the integral $\int_0^\infty \dfrac{\sin ax}{x(x^2+b^2)}dx, a > 0, b > 0$ by the use of Cauchy residue theorem. **[8]**

Q.4 (a) Find the Laurent's expansion for $f(z) = \dfrac{7z-2}{z^3 - z^2 - 2z}$ in the region $1 < |z+1| < 3$ **[6]**

(b) Find the complete solution of $\dfrac{d^2y}{dx^2} - 2\dfrac{dy}{dx} + 2y = x + e^x \cos x$. **[8]**

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Q.5 (a) **[7]**

Find the solution of $x^2\dfrac{d^2y}{dx^2}+x\dfrac{dy}{dx}+y=\log x.\sin(\log x)$.

(b) **[7]**

Find the solution of $\dfrac{d^2y}{dx^2}-6\dfrac{dy}{dx}+9y=\dfrac{e^{3x}}{x^2}$ by the method of variation of parameters.

Q.6 (a) Test the convergence of the series $\sum(1+\frac{1}{n})^{-n2}$. **[7]**

(b) Test the convergence of the series $\dfrac{a}{b}+\dfrac{a+x}{b+x}+\dfrac{a+2x}{b+2x}+\dfrac{a+3x}{b+3x}+.....$ **[7]**

Q.7 (a) Find Laplace transform of t $\cos_{at}$. **[6]**

(b) Evaluate the integral $\int_0^\infty t\,e^{-3t}\sin t\,dt$, by the use of Laplace transform. **[8]**

Q.8 (a) Find the inverse Laplace transform of $\tan^{-1}\left(\dfrac{2}{s^2}\right)$. **[6]**

(b) Find the solution of the following initial value problem by the use of Laplace transform. **[8]**

$$\frac{d^2y}{dt^2}+t\frac{dy}{dt}-y=0,\text{ if } y(0)=0 \text{ and}\left(\frac{dy}{dt}\right)_{t=0}=1.$$

Q.9 (a) Find the Fourier series expansion of the following function. **[7]**

$$f(x)=\begin{cases}0, & -\pi\le x\le 0\\ \sin x, & 0\le x<\pi.\end{cases}$$

Hence deduce that $\dfrac{1}{1.3}-\dfrac{1}{3.5}+\dfrac{1}{5.7}-\dfrac{1}{7.9}+.....=\dfrac{\pi-2}{4}$.

(b) Find the Fourier expansion of $x\sin x$ as Fourier cosine series in the interval $[0,\ \pi]$ **[7]**

. Hence, find the value of $\dfrac{1}{1.3}-\dfrac{1}{3.5}+\dfrac{1}{5.7}-\dfrac{1}{7.9}+.....$

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